ICHTHYOSIS

AT A GLANCE

A heterogeneous group of skin diseases characterized by generalized scaling and, often, areas of thickened skin.
Most types are inherited and usually present at birth; however, some forms are acquired.
Scales may vary in size, color, and distribution across the body.
May be accompanied by erythema, abnormalities of the skin adnexa, and other cutaneous findings.
Can be associated with systemic manifestations such as failure to thrive, increased susceptibility to infection, atopic dermatitis, neurosensory deafness, and neurologic disorders.
Histopathology is typically non-specific, except in conditions such as epidermolytic hyperkeratosis, neutral lipid storage disease, Refsum disease, and acquired ichthyosis associated with sarcoidosis.

OVERVIEW

Early reports of ichthyosis in Indian and Chinese medical literature date back several hundred years B.C. The condition was later described by Robert Willan in 1808.

Ichthyosis may present at birth or develop later in life. It can be limited to the skin or occur in association with abnormalities in other organ systems. While several well-defined types of ichthyosis can be reliably diagnosed based on clinical and genetic features, the significant clinical heterogeneity and the influence of environmental factors on scaling can make specific diagnosis challenging in some patients and families.

Siemens introduced genetic concepts into the classification of ichthyoses, laying the foundation for modern understanding.

Advances in genetic research have identified the underlying gene defects in many forms of ichthyosis—classified as genodermatoses. A new classification system is emerging, based on molecular and genetic etiologies. Identifying the mutated gene provides insight into the pathophysiologic mechanism, enabling more rational approaches to diagnosis, classification, and treatment.

Grouping these disorders by inheritance pattern (Table 47-1) and by underlying molecular abnormality (Table 47-2) enhances understanding of their clinical phenotypes and disease mechanisms. However, further research is needed to fully elucidate how specific gene mutations and protein disruptions lead to clinical disease, and to develop targeted, innovative therapies.

TABLE 47-1

Features of Selected Ichthyoses

DIAGNOSIS	ONSET	CHARACTERISTIC CLINICAL FEATURES	ASSOCIATED FEATURES	GENE	PROTEIN	FUNCTION	SKIN HISTOPATHOLOGY
Autosomal semidominant							
Ichthyosis vulgaris (OMIM #146700)	Infancy/ childhood	Fine or centrally adherent scales with superficial fissuring; relative sparing of flexural areas; more severe on lower extremities; hyperlinear palms and soles	Keratosis pilaris; atopy	*FLG*	Absence of filaggrin	Uncertain	Hyperkeratosis; reduced or absent granular layer
Autosomal dominant							
EHK (bullous CIE; OMIM #113800)	Birth	Heterogeneous presentation					